BLOCKCHAIN TECHNOLOGY

30 Selected Questions — Questions Only

UNIT I — BLOCKCHAIN FUNDAMENTALS

1. Explain any three important properties of blockchain technology and their significance.
2. Differentiate between a centralized database and a blockchain with respect to data management, control, and security.
3. Explain the structure of a blockchain block with a neat diagram, showing the block header, transaction data, and previous block hash.
4. What is block mining? Explain the major steps involved in mining a block in a Proof-of-Work blockchain.
5. Discuss any four common myths about blockchain technology and explain the actual limitations or conditions associated with each.
6. Explain the difference between a public blockchain and a private blockchain, highlighting their access control and use cases.

UNIT II — CRYPTOGRAPHY

7. Differentiate between private-key cryptography and public-key cryptography with suitable examples.
8. Define a cryptographic hash function and explain any four important properties required for blockchain security.
9. Explain the construction of a Merkle tree for four transactions with a neat diagram. What is the significance of the Merkle root?
10. In an RSA system, let $p = 5$, $q = 11$, and the public exponent $e = 3$. Calculate n, $\phi(n)$, and the private exponent d. Show the necessary steps.
11. Explain the process of creating and verifying a digital signature using public-key cryptography.
12. Write down the properties of message digest in detail.
13. Explain how a hash pointer links blocks and contributes to the tamper-evident nature of a blockchain.

UNIT III — DISTRIBUTED SYSTEMS AND CRYPTOCURRENCY

14. Define a distributed system. Explain how blockchain technology uses the principles of distributed systems.
15. Discuss the major design goals of cryptocurrencies.
16. Differentiate between desktop wallets, mobile application-based wallets, and browser-based wallets.
17. What is a decentralized application (DApp)? Compare DApps with conventional applications in terms of architecture, control, and data management.
18. Explain the role of nodes in a peer-to-peer blockchain network and how they maintain a shared ledger.

UNIT IV — CONSENSUS, ETHEREUM, AND HYPERLEDGER

19. Explain the Byzantine Generals Problem and its relevance to blockchain consensus mechanisms.

20. Explain Nakamoto consensus and the role of Proof of Work in selecting the accepted blockchain.
21. Compare Proof of Stake, Proof of Activity, Proof of Burn, Proof of Authority, Proof of Elapsed Time, and Proof of Importance.
22. Explain the role of Ethereum in blockchain technology. How does it support decentralized applications?
23. What is a smart contract? Explain how smart contracts are developed using Solidity and executed on Ethereum.
24. Explain the working of Practical Byzantine Fault Tolerance (PBFT), describing its major phases.
25. Explain the architecture of Hyperledger Fabric. Discuss the roles of peers, ordering services, organizations, and chaincode in a permissioned blockchain.
26. Explain the difference between permissioned and permissionless consensus mechanisms, with suitable examples.

UNIT V — BLOCKCHAIN APPLICATIONS

27. How is blockchain applied in land registry records? Explain in detail.
28. Discuss four potential governmental applications of blockchain technology.
29. Explain how blockchain technology can facilitate cross-border payments. Discuss its advantages and limitations.
30. Explain how blockchain can be used for supply-chain tracking and product provenance. Discuss its benefits and challenges.